

Oobleck

THE SOLID-LIQUID MYSTERY (SESSION 1: THE RULE BREAKER FLUID)

Magic Mud or Science?

Vivaan and Diya found a bowl of strange "magic mud" in their backyard. When they tried to punch it, it felt hard like a rock! But when they let it sit in their hands, it dripped away like water. Is it a solid? Is it a liquid? Let's solve the mystery today!

Your Science Lab Goals

Today, we will become "Mad Scientists" to achieve these goals:

- ✓ **Understand Matter:** Review the basic rules of solids and liquids.
- ✓ **Meet the Rule Breaker:** Discover why Oobleck is a "Non-Newtonian" fluid.
- ✓ **The Pressure Test:** Learn how pressure changes the state of Oobleck.
- ✓ **Make Your Own Oobleck:** Use simple kitchen ingredients to create magic mud!

THE BIG QUESTION

If you could walk on a pool of liquid without sinking, would you try it? Oobleck makes that possible!

Solids and Liquids: The Normal Rules

Most things in our world follow simple rules. Let's look at what we see in our Indian homes:

1. Solids (The Firm Ones)

Solids hold their shape no matter what. A Stainless Steel Glass is a solid. It doesn't drip or flow, and it stays hard when you touch it.

2. Liquids (The Flowy Ones)

Liquids don't have their own shape. They take the shape of whatever they are in. Mango Lassi or Coconut Oil are liquids. They flow, drip, and pour easily.

Oobleck: The Rule Breaker!

Oobleck is a special substance that doesn't like following the normal rules of solids and liquids. It is a **Non-Newtonian Fluid**.

The Mood-Changing Fluid: Oobleck changes its "thickness" based on how you treat it. If you are rough with it, it acts like a solid. If you are gentle, it acts like a liquid!

The Pressure Secret:

- ☑ **High Pressure:** When you press hard or punch it quickly, it feels firm like a Cricket Bat or a brick!
- ☑ **Low Pressure:** When you let it sit softly in your palm, it oozes and drips like warm honey or oil!

The Oobleck Lab: Making Magic Mud

You can make this amazing substance right in your kitchen!

Ingredients:

- 2 Cups of Cornflour (Makai ka atta).
- 1 Cup of Water.
- *Optional:* A few drops of your favorite food colour!

How to Mix:

1. Pour the cornflour into a large bowl.
2. Slowly add the water while mixing with your hands.
3. Keep mixing until it feels like a thick paste that "tears" when you pull it but "melts" when you stop.

Experiment: The Pressure Test

Now that you've made your Oobleck, it's time to test its magic powers!

Exhibit 1: The Punch Test

Make a fist and punch the Oobleck surface quickly. What happens? Your fist won't sink! It will bounce off as if you hit a solid wall.

Exhibit 2: The Ooze Test

Now, gently place your open hand on top of the Oobleck. What happens? Your hand will slowly sink deep into the goo, just like it's in water!

The Science Secret

Why does Oobleck behave so strangely? It's all about how the tiny pieces of cornflour interact with water.

The "Tangled" Particles:

Inside Oobleck, the cornflour bits are like tiny grains of sand.

- When you push fast, the water gets squeezed out from between them, and the grains "lock" together like a solid.
- When you go slow, the water can flow around the grains, allowing them to slide past each other like a liquid!

SCIENCE ARTIST IDEA:

Try making a "ball" of Oobleck by rolling it quickly between your palms. As soon as you stop rolling, watch it "melt" back into a puddle in your hand!

The Mad Scientist's Clean-up Guide

Science is fun, but a good scientist always cleans up their lab!

IMPORTANT: PROTECT YOUR PIPES!

NEVER pour Oobleck down the kitchen or bathroom sink. As it settles, it can turn solid and clog the pipes completely!

How to Clean Up:

- **The Bin:** Scrape your Oobleck into the trash or a compost bin.
- **Dry & Wipe:** Let any spills on the table dry out. They will turn back into white powder that you can easily wipe or vacuum up.
- **Hand Wash:** Rinse your hands over a bowl of water first, then pour that water outside or in the trash before using the sink for a final wash.

Interactive: Solid, Liquid, or Both?

Check the box for each item based on its state of matter!

ITEM	SOLID	LIQUID	BOTH!
Steel Ladle	☐	☐	☐
Water	☐	☐	☐
Oobleck	☐	☐	☐
Cricket Ball	☐	☐	☐

Lesson Recap

You have successfully solved the Solid-Liquid Mystery! Let's remember our discoveries:

- **Non-Newtonian:** Oobleck is a fluid that doesn't follow normal rules.
- **Pressure:** It turns hard when you hit it and runny when you hold it.
- **Ingredients:** You only need cornflour and water to make it!
- **Safety:** Always throw Oobleck in the trash, never down the sink!

Stay Messy, Stay Curious!